Response to the Major-Revision Examiner Report

Gaussian-Process Reconstruction of the Mapping-QCLE Excess Term: A Moving-Cloud Formulation and Failure Analysis

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Revision overview

The revision presents the work as a controlled negative-result study. Accurate density interpolation is separated from operator fidelity, SEO structure, raw conservation, stochastic stability, and physical accuracy. Because these requirements are not jointly satisfied, the revised thesis does not call the MIDPOINT construction reliable and does not claim a systematic improvement over PBME.

Final mandatory and requested corrections

1 — Time-step inventory reconciled

Status: Implemented and verified in the revised source

Correction: The campaign is stated as 24 paired momentum–seed–step configurations, 48 individual method executions, and four independent seeds.

Final location: Thesis p. 80, Section 5 time-step tests; Table G.4.

Archive evidence: final_reviewer_closure/timestep/timestep_run_by_run.csv

2 — Reference settings synchronized

Status: Implemented and verified in the revised source

Correction: Table 6.12 is generated from the same full-precision CSVs as Table F.1 and lists momentum, final time, all domain and grid/time levels, physical edge statistic, and a defined CFL ratio.

Final location: Thesis p. 104, Table 6.12; Appendix F, Table F.1.

Archive evidence:
final_reviewer_closure/reference_settings_by_method_and_momentum.csv

3 — Temporal orders corrected

Status: Implemented and verified in the revised source

Correction: The text assigns approximately second-order temporal behaviour to TDSE and approximately fourth-order temporal behaviour to grid-QCLE RK4; spatial/grid behaviour is treated separately.

Final location: Thesis p. 101, Section 6 reference controls; Tables 6.13–6.14.

Archive evidence: final_reviewer_closure/reference_tdse/tdse_three_level.csv and reference_grid_qcle/qcle_three_level.csv

4 — Numerical-noise and stochastic guards enforced

Status: Implemented and verified in the revised source

Correction: The absolute-plus-relative floor is 1e-12 + 1e-12 times the maximum level scale. Noise-limited, nonmonotone, and rapidly contracting nonasymptotic rows are rejected; for both stochastic moving-cloud methods, PBME and MIDPOINT, both differences must also exceed pooled independent-seed variation.

Final location: Thesis p. 80, Eq. (declared numerical-noise floor); Tables 6.7, 6.13, 6.14, and G.4.

Archive evidence: full-precision order-reason and verdict columns in the three reference/time-step CSVs

5 — Abstract claim calibrated

Status: Implemented and verified in the revised source

Correction: The abstract now says numerically controlled reference solutions and does not call them converged solutions.

Final location: Thesis p. i, Abstract.

Archive evidence: Thesis/Thesis.tex

6 — Independent clouds no longer called convergent

Status: Implemented and verified in the revised source

Correction: N=500, 1000, and 2000 are described as independently sampled, nonnested cloud enlargement; no deterministic cloud order is claimed.

Final location: Thesis p. 93, Section 6 independent-cloud enlargement; Table 6.8.

Archive evidence: final_reviewer_closure/support/independent_cloud_summary.csv

7 — Retrievable release record supplied

Status: Implemented and verified in the revised source

Correction: The final release identifies the archive filename, public release URL, final release tag, frozen source/evidence commit, archive SHA-256, checksum-index SHA-256, environment, manifests, clean-room verification record, and reproduction instructions. It also states that no DOI or institutional persistent identifier is assigned.

Final location: Thesis p. 170, Appendix G, Section G.5 archive record.

Archive evidence: frozen_numerical_evidence_payload.zip;
`https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01-r3`

8 — Visible typography repaired

Status: Implemented and verified in the revised source

Correction: The chapter range and radial-basis cross-validation phrase are kept unbroken; the dense Appendix G evidence remains complete.

Final location: Thesis Chapter 1 roadmap and Chapter 4 cross-validation text.

Archive evidence: Thesis/Thesis.tex

9 — Final response crosswalk added

Status: Implemented and verified in the revised source

Correction: The ten gates and all 48 I/M/L items are generated from a 58-row matrix with a unique correction, exact section/page locator, and named evidence artifact for every row.

Final location: This response, Ten acceptance gates and I/M/L sections.

Archive evidence: reviewer_data_audit/response_item_audit.csv

10 — Shared stochastic refinement rule made consistent

Status: Implemented and verified in the revised source

Correction: Section 6.4 now states that both PBME and MIDPOINT must clear the same numerical-noise and pooled independent-seed-dispersion guards before any time-step contraction is interpreted.

Final location: Thesis p. 89, Section 6.4; Table 6.7.

Archive evidence: final_reviewer_closure/timestep/timestep_run_by_run.csv

11 — Signed central second moments defined and relabelled

Status: Implemented and verified in the revised source

Correction: The archived R_var and P_var fields are defined mathematically as signed central second moments and displayed as M2,signed rather than variances. Negative values are classified as physical-admissibility failures.

Final location: Thesis p. 142, Appendix E physical observables; Tables 6.7–6.9 and 6.13–6.14.

Archive evidence: final_reviewer_closure/support/independent_cloud_run_by_run.csv; replication/four_seed_summary.csv

12 — Cloud-size verdict hierarchy enforced

Status: Implemented and verified in the revised source

Correction: Cloud-size verdicts now reject numerical-noise-limited changes first, reject nonfinite or physically inadmissible outputs second, and compare only remaining resolvable changes with seed dispersion.

Final location: Thesis p. 93, Section 6.5 and Table 6.8.

Archive evidence: final_reviewer_closure/support/cloud_size_verdict_audit.csv

13 — Low-momentum grid-QCLE role reclassified

Status: Implemented and verified in the revised source

Correction: A declared observable-specific finest-grid tolerance and usable Richardson-style error estimate are now reported. Because six of eight P_init=20 rows fail the tolerance, that case is a numerical-sensitivity reference rather than a resolved accuracy standard.

Final location: Thesis p. 101, Section 6.7 and Table 6.14.

Archive evidence: final_reviewer_closure/reference_grid_qcle/qcle_reference_accuracy.csv

14 — Spatial exponents no longer called method orders

Status: Implemented and verified in the revised source

Correction: Spatial/grid values are labelled three-level effective contraction exponents. The p<=6 ceiling is identified as a conservative predeclared screen, not a theoretical expected method order.

Final location: Thesis p. 101, Section 6.7; Tables 6.13–6.14.

Archive evidence: final_reviewer_closure/reference_tdse/tdse_three_level.csv; reference_grid_qcle/qcle_three_level.csv

15 — Four-seed interval interpretation qualified

Status: Implemented and verified in the revised source

Correction: The paired Student-t intervals are described as sensitivity summaries only. Table 6.15 reports whether the four raw paired signs are consistent, with the individual Appendix G values primary.

Final location: Thesis p. 112, Table 6.15 and Figure 6.5; Appendix G per-seed tables.

Archive evidence:
final_reviewer_closure/physical_comparison/paired_improvement_summary.csv; observable_errors_method_pair_by_seed.csv; density_errors_method_pair_by_seed.csv

16 — Formal-order scope narrowed

Status: Implemented and verified in the revised source

Correction: The formal statement is restricted to a smooth fixed-representation semi-discrete midpoint system; the moving-support refit and safe-profile regularity assumptions are explicitly not established.

Final location: Thesis p. 75, Section 5.5 and Chapter 5 summary.

Archive evidence: Thesis/Thesis.tex

17 — Current response included in the immutable release

Status: Implemented and verified in the revised source

Correction: The newly compiled response PDF, source, and updated crosswalk are release assets alongside the thesis, rather than being omitted from the reviewer-delivery package.

Final location: This response and the release asset manifest.

Archive evidence: Reviewer_Response.pdf; Reviewer_Response.tex; GitHub release manifest

Ten acceptance gates

Gate 1

Status: Addressed in the revised thesis

Exact correction: Gate 1 closure: Each chapter states its question, motivation, approach, and outcome; Chapter 6 opens with the decisive reliability question and controlled negative answer.

Thesis locator: Section 1.9, printed p. 25; chapter openings/closings at printed pp. 27, 39, 49, 65, 83, and 117–119.

Evidence artifact: Thesis/Thesis.tex; Thesis/Chapter6_VerifiedResults.tex; Thesis/Chapter7_Conclusions.tex

Gate 2

Status: Addressed in the revised thesis

Exact correction: Gate 2 closure: The novelty claim is literature-bounded and restricted to the application-specific product-GP/MIDPOINT construction and its failure analysis.

Thesis locator: Section 1.9, printed p. 25; Section 7.1, printed p. 117.

Evidence artifact: Thesis/Thesis.tex; Thesis/Chapter7_Conclusions.tex and cited literature

Gate 3

Status: Addressed in the revised thesis

Exact correction: Gate 3 closure: All retained figures define their quantities, aggregation, uncertainty, normalization, sign convention, and interpretation and are hash-linked to source CSVs.

Thesis locator: Figures 6.1–6.5, printed pp. 87, 88, 99, 102, and 115.

Evidence artifact: final_reviewer_closure/figures/FIGURE_DATA_CROSSWALK.csv

Gate 4

Status: Addressed in the revised thesis

Exact correction: Gate 4 closure: All three manufactured cloud-pair seeds are printed for every N, regularization, support type, and density/gradient/operator metric; the Chapter 6 summaries retain the mean and sample SD.

Thesis locator: Section 6.2, printed pp. 84–86; Tables 6.2–6.4 and Figure 6.1; Appendix G, Tables G.1–G.3, printed pp. 151–156.

Evidence artifact: final_reviewer_closure/manufactured/manufactured_complete.csv; manufactured_summary.csv; manufactured_refinement_verdicts.csv

Gate 5

Status: Addressed in the revised thesis

Exact correction: Gate 5 closure: Production and reference orders are reconstructible and guarded. For both PBME and MIDPOINT, both differences must exceed numerical noise and pooled independent-seed dispersion; reference time/grid controls remain separate.

Thesis locator: Sections 5.9.5, 6.4, and 6.7, printed pp. 80, 89–93, and 101–110; Tables 6.7 and 6.12–6.14; Table G.4, printed pp. 158–160.

Evidence artifact: final_reviewer_closure/timestep/timestep_run_by_run.csv; reference_tdse/tdse_three_level.csv; reference_grid_qcle/qcle_three_level.csv

Gate 6

Status: Addressed in the revised thesis

Exact correction: Gate 6 closure: N=500, 1000, and 2000 use independent nonnested clouds. Verdicts apply numerical-noise rejection, then physical-admissibility rejection, and only then a seed-dispersion comparison; no deterministic convergence is claimed.

Thesis locator: Section 4.7.5, printed p. 62; Section 6.5, printed pp. 93–97; Table 6.8.

Evidence artifact: final_reviewer_closure/support/independent_cloud_summary.csv; support/cloud_size_verdict_audit.csv

Gate 7

Status: Addressed in the revised thesis

Exact correction: Gate 7 closure: Physical comparisons are reported as per-seed absolute errors and paired MIDPOINT-minus-PBME differences. Four-seed Student-t intervals are descriptive only, and raw per-seed sign consistency is stated explicitly.

Thesis locator: Section 6.8, printed pp. 111–115; Table 6.15 and Figure 6.5; Appendix G, Tables G.7–G.8, printed pp. 167–168.

Evidence artifact:
final_reviewer_closure/physical_comparison/density_errors_method_pair_by_seed.csv; observable_errors_method_pair_by_seed.csv; paired_improvement_summary.csv

Gate 8

Status: Addressed in the revised thesis

Exact correction: Gate 8 closure: The archive record distinguishes the final release tag, which identifies the tagged document commit, from the frozen source/evidence commit and states that the GitHub release is versioned public access, not a DOI or institutional persistent identifier.

Thesis locator: Appendix G.5, printed p. 170; this response Availability note.

Evidence artifact: release `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01-r3`; frozen_numerical_evidence_payload.zip; frozen source/evidence commit `1f170df258cb4312 3af1a040a9973ca6996c804b`; archive SHA-256 `9752f3e16a305225ed30796d799d5a9795d09d8c debd1ee74648054cf84af87e`; checksum-index SHA-256 `e4749382e8ac55e1b20f129e2c80402b 6ad112665bbd1a00be60ed6075b223b2`; FINAL_RUN_MANIFEST.json; FIGURE_DATA_CROSSWALK.csv; table_data_crosswalk.csv; environment.json; README.md; CLEAN_ROOM_VERIFICATION.json

Gate 9

Status: Addressed in the revised thesis

Exact correction: Gate 9 closure: The exact title is 'Gaussian-Process Reconstruction of the Mapping-QCLE Excess Term: A Moving-Cloud Formulation and Failure Analysis' in the title page, PDF metadata, response heading/metadata, citation record, repository README, and release title.

Thesis locator: Thesis title page and PDF metadata; this response title and PDF metadata; public repository/release title.

Evidence artifact: Thesis/Thesis.tex; Thesis/Thesis.pdf metadata; Reviewer_Response.tex; CITATION.cff; GitHub release metadata

Gate 10

Status: Addressed in the revised thesis

Exact correction: Gate 10 closure: The response is now generated from a 58-row gate/item audit matrix; every row has a unique correction, exact section/page locator, and named evidence artifact.

Thesis locator: This response, Ten acceptance gates and I/M/L sections.

Evidence artifact: reviewer_data_audit/response_item_audit.csv

I-items

I-1 — Complete three-seed manufactured result not shown

Status: Addressed in the revised thesis

Exact correction: All three manufactured cloud-pair seeds are printed for every N, regularization, support type, and density/gradient/operator metric; the Chapter 6 summaries retain the mean and sample SD.

Thesis locator: Section 6.2, printed pp. 84–86; Tables 6.2–6.4 and Figure 6.1; Appendix G, Tables G.1–G.3, printed pp. 151–156.

Evidence artifact: final_reviewer_closure/manufactured/manufactured_complete.csv; manufactured_summary.csv; manufactured_refinement_verdicts.csv

I-2 — Time-step order conclusion cannot be reconstructed

Status: Addressed in the revised thesis

Exact correction: Each method/momentum/seed/observable row reports the three endpoint values, both aligned time-series differences, the numerical floor, pooled seed spread, guarded order, verdict, and reason.

Thesis locator: Sections 5.9.5 and 6.4, printed pp. 80 and 89–93; Table 6.7; Appendix G, Table G.4, printed pp. 158–160.

Evidence artifact: final_reviewer_closure/timestep/timestep_run_by_run.csv

I-3 — Reference-convergence orders are unassigned

Status: Addressed in the revised thesis

Exact correction: TDSE and grid-QCLE time and spatial/grid ladders are separated; orders are shown only after the declared noise, monotonicity, and asymptotic guards pass.

Thesis locator: Section 6.7, printed pp. 101–110; Tables 6.12–6.14; Appendix F.1, printed pp. 145–147.

Evidence artifact: final_reviewer_closure/reference_tdse/tdse_three_level.csv; reference_grid_qcle/qcle_three_level.csv

I-4 — MIDPOINT central value at P0=20 absent

Status: Addressed in the revised thesis

Exact correction: The MIDPOINT P_init=20 central values are reported from seeds 11, 29, 47, and 73 with mean, sample SD, spread, and Student-t interval.

Thesis locator: Section 6.5, printed pp. 93–97; Table 6.9 and Figure 6.3; corresponding dt=0.25 rows of Table G.4, printed pp. 158–160.

Evidence artifact: final_reviewer_closure/replication/four_seed_values.csv; four_seed_summary.csv

I-5 — Identical-support pass incompletely reported

Status: Addressed in the revised thesis

Exact correction: The identical-support KDE/GP comparison uses the same PBME cloud, weights, bandwidth, normalization, and grid; only the declared E1 value-reconstruction gate is evaluated.

Thesis locator: Section 6.3, printed pp. 86–89; Table 6.6, printed p. 88.

Evidence artifact: final_reviewer_closure/preserved_evidence/kde_gp_baseline.csv

I-6 — 41 retained figures have no complete provenance

Status: Addressed in the revised thesis

Exact correction: Every retained figure now defines the plotted quantity, normalization, seed aggregation, uncertainty display, sign convention, and evidentiary limit in its caption.

Thesis locator: Figures 6.1–6.5, printed pp. 87, 88, 99, 102, and 115.

Evidence artifact: final_reviewer_closure/figures/FIGURE_DATA_CROSSWALK.csv and the five hashed source CSV entries named there

I-7 — Figure-dependent descriptions not interpretable

Status: Addressed in the revised thesis

Exact correction: All qualitative legacy comparisons were removed; the five retained figures are paired with quantitative tables and explicit metrics.

Thesis locator: Sections 6.2–6.8, printed pp. 84–115; Figures 6.1–6.5 and Tables 6.2–6.15.

Evidence artifact: final_reviewer_closure/figures/FIGURE_DATA_CROSSWALK.csv; table_data_crosswalk.csv

I-8 — Plotting thresholds undefined

Status: Addressed in the revised thesis

Exact correction: Every acceptance/display threshold is stated numerically at use: the E1 gate, absolute-plus-relative order floor, edge-mass criterion, conservative effective-exponent screen, and tail-mass rule. The p<=6 screen is explicitly not a theoretical method order.

Thesis locator: Sections 5.9.5 and 6.3–6.7, printed pp. 80 and 86–110; Tables 6.6–6.7 and 6.11–6.14.

Evidence artifact: final_reviewer_closure/preserved_evidence/kde_gp_baseline.csv; timestep/timestep_run_by_run.csv; reference_tdse/tdse_three_level.csv; reference_grid_qcle/qcle_three_level.csv; tail_sensitivity/threshold_sweep.csv

I-9 — Diagnostics divide by undocumented |y0| tail

Status: Addressed in the revised thesis

Exact correction: The tail analysis reports threshold, excluded point fraction, excluded absolute physical mass, normalization, energy, and signed/absolute ESS; no nontrivial negligible-mass plateau is claimed.

Thesis locator: Section 6.6, printed pp. 98–102; Table 6.11 and Figure 6.4; Appendix G, Tables G.5–G.6, printed pp. 162–166.

Evidence artifact: final_reviewer_closure/tail_sensitivity/y0_distribution_paired.csv; threshold_sweep.csv

I-10 — 'Applied source after stabilization' unspecified

Status: Addressed in the revised thesis

Exact correction: The reported MIDPOINT branch applies Q_applied=Q_raw: apply_q_clip=false and omega_clip_quantile=null; clipped and renormalized alternatives are not substituted.

Thesis locator: Sections 5.2 and 5.4.2, printed pp. 67–74; Section 6.6, Table 6.10, printed p. 98.

Evidence artifact: Dynamics.py; final_reviewer_closure/FINAL_RUN_MANIFEST.json; preserved_evidence/raw_conservation.csv

I-11 — 'Physically implausible' analytic observables not reported

Status: Addressed in the revised thesis

Exact correction: The unsupported analytic-GP physical-observable assertion was removed; no uncomputed analytic-observable table or conclusion remains.

Thesis locator: Sections 6.8–6.9 and 7.3, printed pp. 111–118; retained physical claims are limited to Tables 6.15 and G.7–G.8.

Evidence artifact: Thesis/Thesis.tex; final_reviewer_closure/physical_comparison/paired_improvement_summary.csv

I-12 — 'Only weakly' undefined and conflicts with Table 6.6

Status: Addressed in the revised thesis

Exact correction: The former word 'weakly' was replaced by estimator- and momentum-specific four-cloud statistics; four clouds, not trajectories, define the independent sample size.

Thesis locator: Section 6.5, printed pp. 93–97; Table 6.9 and Figure 6.3.

Evidence artifact: final_reviewer_closure/replication/four_seed_values.csv; four_seed_summary.csv

I-13 — High-momentum agreement not quantitative

Status: Addressed in the revised thesis

Exact correction: Physical comparisons are reported as per-seed absolute errors and paired MIDPOINT-minus-PBME differences. Four-seed Student-t intervals are descriptive only, and raw per-seed sign consistency is stated explicitly.

Thesis locator: Section 6.8, printed pp. 111–115; Table 6.15 and Figure 6.5; Appendix G, Tables G.7–G.8, printed pp. 167–168.

Evidence artifact:
final__reviewer__closure/physical__comparison/density__errors__method__pair__by__seed.csv; observable__errors__method__pair__by__seed.csv; paired__improvement__summary.csv

I-14 — Exact reference settings outside the thesis

Status: Addressed in the revised thesis

Exact correction: The thesis prints every reference setting and now gives an explicit finest-grid tolerance and usable Richardson-style estimate. The P__init=20 grid-QCLE case is classified only as a numerical-sensitivity reference because six of eight observables miss tolerance.

Thesis locator: Section 6.7, printed pp. 101–110; Tables 6.12–6.14; Appendix F.1–F.3, printed pp. 145–149.

Evidence artifact:
final__reviewer__closure/reference__settings__by__method__and__momentum.csv; reference__tdse/tdse__three__level.csv; reference__grid__qcle/qcle__three__level.csv; reference__grid__qcle/qcle__reference__accuracy.csv

I-15 — Immutable configuration / versioned record not identified

Status: Addressed in the revised thesis

Exact correction: The archive record distinguishes the final release tag, which identifies the tagged document commit, from the frozen source/evidence commit and states that the GitHub release is versioned public access, not a DOI or institutional persistent identifier.

Thesis locator: Appendix G.5, printed p. 170; this response Availability note.

Evidence artifact: release `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01-r3`; frozen__numerical__evidence__payload.zip; frozen source/evidence commit `1f170df258cb43123af1a040a9973ca6996c804b`; archive SHA-256 `9752f3e16a305225ed30796d799d5a9795d09d8cdebd1ee74648054cf84af87e`; checksum-index SHA-256 `e4749382e8ac55e1b20f129e2c80402b6ad112665bbd1a00be60ed6075b223b2`; FINAL__RUN__MANIFEST.json; FIGURE__DATA__CROSSWALK.csv; table__data__crosswalk.csv; environment.json; README.md; CLEAN__ROOM__VERIFICATION.json

I-16 — Thesis title not uniquely identified across artifact

Status: Addressed in the revised thesis

Exact correction: The exact title is 'Gaussian-Process Reconstruction of the Mapping-QCLE Excess Term: A Moving-Cloud Formulation and Failure Analysis' in the title page, PDF metadata, response heading/metadata, citation record, repository README, and release title.

Thesis locator: Thesis title page and PDF metadata; this response title and PDF metadata;

public repository/release title.

Evidence artifact: Thesis/Thesis.tex; Thesis/Thesis.pdf metadata; Reviewer_Response.tex; CITATION.cff; GitHub release metadata

M-items

M-1 — Contribution not separated from known mathematical fact

Status: Addressed in the revised thesis

Exact correction: The thesis explicitly says that nonidentifiability of ambient normal derivatives is known; originality is restricted to constructing and testing this application-specific product-GP/MIDPOINT chain.

Thesis locator: Section 7.1, printed p. 117.

Evidence artifact: Thesis/Chapter7_Conclusions.tex; cited derivative-GP and probabilistic-numerics literature

M-2 — 'Projected density is represented' is misleading

Status: Addressed in the revised thesis

Exact correction: The projected density is the physical target, while the tested product surrogate is not projection-enforcing; measured SEO leakage is therefore a failure diagnostic.

Thesis locator: Sections 4.1.1–4.1.2, printed pp. 50–51; Section 6.3, printed pp. 86–89; Table 6.5 and Figure 6.2.

Evidence artifact: final_reviewer_closure/preserved_evidence/projection_leakage.csv

M-3 — 'Semi-discretization of the complete generator' too strong

Status: Addressed in the revised thesis

Exact correction: The method is described as one attempted moving-cloud collocation of the formal Hamiltonian and excess-source terms, not a validated semi-discretization of the complete generator.

Thesis locator: Sections 5.2–5.4, printed pp. 67–74; Chapter 6 opening, printed p. 83; Section 7.3, printed p. 118.

Evidence artifact: Thesis/Thesis.tex; Thesis/Chapter6_VerifiedResults.tex; Thesis/Chapter7_Conclusions.tex

M-4 — 'Corrected method' implies correctness

Status: Addressed in the revised thesis

Exact correction: Method-level 'corrected method/evolution' language was replaced by 'MIDPOINT prototype', 'source update', or 'tested excess-source branch'.

Thesis locator: Chapters 5–7, printed pp. 65–119.

Evidence artifact: Thesis/Thesis.tex; final_acceptance_check.py forbidden-phrase audit

M-5 — 'Four checks form a controlled argument' overstates

Status: Addressed in the revised thesis

Exact correction: The checks are described as evidence of failure under tested settings; the thesis lists untested transferability and states that causal allocation is nonunique.

Thesis locator: Sections 7.2–7.5, printed pp. 117–118.

Evidence artifact: Thesis/Chapter7_Conclusions.tex; Tables 6.2–6.15

M-6 — 'Support refinement/convergence' used too freely

Status: Addressed in the revised thesis

Exact correction: N=500, 1000, and 2000 use independent nonnested clouds. Verdicts apply numerical-noise rejection, then physical-admissibility rejection, and only then a seed-dispersion comparison; no deterministic convergence is claimed.

Thesis locator: Section 4.7.5, printed p. 62; Section 6.5, printed pp. 93–97; Table 6.8.

Evidence artifact: final_reviewer_closure/support/independent_cloud_summary.csv; support/cloud_size_verdict_audit.csv

M-7 — l2=0.01 control does not globally rule out regularization

Status: Addressed in the revised thesis

Exact correction: The 0.01 result is restricted to its pilot policy and tested cells; all three regularizations are compared without claiming that regularization globally cures instability.

Thesis locator: Section 4.7.5, printed p. 62; Section 6.2, printed pp. 84–86; Tables 6.2–6.4 and Figure 6.1.

Evidence artifact: final_reviewer_closure/manufactured/manufactured_complete.csv; manufactured_summary.csv

M-8 — Formal second order is not demonstrated order

Status: Addressed in the revised thesis

Exact correction: The formal statement is restricted to a smooth fixed-representation semi-discrete midpoint system; moving-support refit and safe-profile assumptions are not established. Neither PBME nor MIDPOINT receives an observed production order unless both numerical-noise and pooled seed-dispersion guards pass.

Thesis locator: Sections 5.5 and 5.9.5, printed pp. 75 and 80; Section 6.4, printed pp. 89–93; Table 6.7 and Table G.4.

Evidence artifact: final_reviewer_closure/timestep/timestep_run_by_run.csv

M-9 — 'Qualitative visualization' does not admit untraceable images

Status: Addressed in the revised thesis

Exact correction: All 41 untraceable legacy figures and dependent conclusions were removed; five figures regenerated from hash-verified CSVs remain.

Thesis locator: Figures 6.1–6.5, printed pp. 87, 88, 99, 102, and 115.

Evidence artifact: final_reviewer_closure/figures/FIGURE_DATA_CROSSWALK.csv

M-10 — Independent shape normalization conceals the failure

Status: Addressed in the revised thesis

Exact correction: Unit-mass shape comparisons are labelled shape-only and are interpreted only after raw normalization, trace, and energy. Signed central second moments are explicitly defined and negative values are physical-admissibility failures, not variances.

Thesis locator: Section 6.6, printed pp. 98–102; Table 6.10 and Figure 6.3; Section 6.8, printed pp. 111–115.

Evidence artifact: final_reviewer_closure/preserved_evidence/raw_conservation.csv; physical_comparison/paired_improvement_summary.csv

M-11 — 'Less than ~1% per step' reads as a pointwise bound

Status: Addressed in the revised thesis

Exact correction: The thesis now states explicitly that a ratio of global RMS norms is a global diagnostic, not a pointwise per-step bound; the former less-than-one-percent wording is not used.

Thesis locator: Section 5.9, printed p. 78.

Evidence artifact: Thesis/Thesis.tex

M-12 — 'Not confined to negligible-label tails' unsupported

Status: Addressed in the revised thesis

Exact correction: Tail attribution is supported only through quantile-resolved labels, threshold sweeps, excluded physical mass, normalization, energy, and ESS; no unique tail cause is claimed.

Thesis locator: Section 6.6, printed pp. 98–102; Table 6.11 and Figure 6.4; Tables G.5–G.6, printed pp. 162–166.

Evidence artifact: final_reviewer_closure/tail_sensitivity/y0_distribution_paired.csv; threshold_sweep.csv

M-13 — Self-normalized GP normalization too prominent

Status: Addressed in the revised thesis

Exact correction: Self-normalized unity is identified as tautological and excluded from conservation evidence; raw integral, trace, energy, population positivity, and signed central-second-moment admissibility are primary.

Thesis locator: Section 6.6, printed pp. 98–102; Table 6.10 and Figure 6.3.

Evidence artifact: final_reviewer_closure/preserved_evidence/raw_conservation.csv

M-14 — No explicit estimator hierarchy

Status: Addressed in the revised thesis

Exact correction: The estimator hierarchy is stated before results: raw observables/invariants, then analytic/reference errors, then internal GP diagnostics.

Thesis locator: Section 6.1, printed pp. 83–84; Table 6.1.

Evidence artifact: final_reviewer_closure/validation_inventory.csv; table_data_crosswalk.csv

M-15 — 'Anchor-cloud estimator is more defensible' too strong

Status: Addressed in the revised thesis

Exact correction: Neither anchor-cloud nor analytic-GP estimator is called validated in the low-signed-ESS regime; low ESS is treated as estimator unreliability.

Thesis locator: Sections 5.9.4 and 6.6, printed pp. 80 and 98–102; Section 7.3, printed p. 118.

Evidence artifact: final_reviewer_closure/tail_sensitivity/threshold_sweep.csv; preserved_evidence/raw_conservation.csv

M-16 — Undefined proximity language for the reference benchmark

Status: Addressed in the revised thesis

Exact correction: Reference proximity is defined only by printed absolute errors. The low-momentum grid-QCLE result is not treated as a resolved accuracy standard, and the decisive conclusion relies primarily on TDSE, raw conservation, SEO projection, and replication.

Thesis locator: Sections 6.7–6.8, printed pp. 101–115; Table 6.15; Tables G.7–G.8, printed pp. 167–168.

Evidence artifact:
final_reviewer_closure/physical_comparison/density_errors_method_pair_by_seed.csv; observable_errors_method_pair_by_seed.csv

M-17 — Causation attributed to the excess update, not the discretization

Status: Addressed in the revised thesis

Exact correction: Failure is attributed to the tested nonprojected, nonconservative product-GP/MIDPOINT discretization, not to the continuum excess term itself.

Thesis locator: Sections 7.2–7.3 and 7.6, printed pp. 117–119.

Evidence artifact: Thesis/Chapter7_Conclusions.tex; preserved_evidence/projection_leakage.csv; raw_conservation.csv

M-18 — 'PBME reconstruction is faithful' broader than the gate

Status: Addressed in the revised thesis

Exact correction: The identical-cloud result is limited to passing the declared E1 value-reconstruction gate and is not generalized to derivatives, propagation, or physical fidelity.

Thesis locator: Section 6.3, printed pp. 86–89; Table 6.6, printed p. 88.

Evidence artifact: final_reviewer_closure/preserved_evidence/kde_gp_baseline.csv

M-19 — Failure attribution not fully isolated

Status: Addressed in the revised thesis

Exact correction: The thesis presents a compatible multi-stage failure pathway and explicitly states that the tests do not uniquely apportion causality.

Thesis locator: Section 7.2, printed p. 117; Section 7.3, printed p. 118.

Evidence artifact: Thesis/Chapter7_Conclusions.tex; manufactured, projection, tail, and raw-conservation CSVs

M-20 — 'Reproducible basis' contradicts archive defects

Status: Addressed in the revised thesis

Exact correction: Reproducibility is bounded to the versioned public GitHub release and its checksum record; the text explicitly states that no DOI or institutional persistent identifier has been assigned.

Thesis locator: Section 7.5, printed p. 118; Appendix G.5, printed p. 170; this response Availability note.

Evidence artifact: `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01-r3`; frozen_numerical_evidence_payload_manifest.json; CLEAN_ROOM_VERIFICATION.json

M-21 — Objective broader than the tested construction

Status: Addressed in the revised thesis

Exact correction: The objective now names one product-GP/moving-cloud/explicit-MIDPOINT construction on a one-dimensional, two-state avoided-crossing benchmark.

Thesis locator: Section 1.9, printed p. 25.

Evidence artifact: Thesis/Thesis.tex

M-22 — Chapter 6 opens with the wrong decision question

Status: Addressed in the revised thesis

Exact correction: Chapter 6 opens with the physical reliability question and its controlled negative answer before presenting any internal diagnostic.

Thesis locator: Chapter 6 opening and Section 6.1, printed pp. 83–84.

Evidence artifact: Thesis/Chapter6_VerifiedResults.tex

M-23 — 'Full-density representation' ambiguous

Status: Addressed in the revised thesis

Exact correction: Ambiguous 'full-density representation' language is absent; the text distinguishes the projected physical target from the unprojected tested product surrogate.

Thesis locator: Sections 4.1.1–4.1.2, printed pp. 50–51; Section 6.3, printed pp. 86–89.

Evidence artifact: Thesis/Thesis.tex;
final_reviewer_closure/preserved_evidence/projection_leakage.csv

M-24 — 'Selected l2 = 0.01' needs qualification

Status: Addressed in the revised thesis

Exact correction: Regularization 0.01 is called pilot-selected; complete operator tests are repeated at 1e-6, 0.01, and production 0.05 over all four N values and three seeds.

Thesis locator: Section 4.7.5, printed p. 62; Section 6.2, printed pp. 84–86; Tables 6.2–6.4 and G.1–G.3.

Evidence artifact: final_reviewer_closure/manufactured/manufactured_complete.csv; manufactured_summary.csv

M-25 — Completion counts confused with validation

Status: Addressed in the revised thesis

Exact correction: Completion counts are confined to the inventory; scientific validation is organized by physical question, controlled variation, independent realizations, and diagnostic.

Thesis locator: Section 6.1, printed pp. 83–84; Table 6.1.

Evidence artifact: final_reviewer_closure/validation_inventory.csv

L-items

L-1 — Use one method vocabulary

Status: Addressed in the revised thesis

Exact correction: One vocabulary is used: PBME baseline, MIDPOINT prototype, product-GP surrogate, excess-source update, and tested discretization; 'corrected method' is absent.

Thesis locator: Sections 1.9, 5.1–5.4, and 6.1, printed pp. 25, 66–74, and 83–84.

Evidence artifact: Thesis/Thesis.tex; final_acceptance_check.py terminology audit

L-2 — Define every estimator at first use and map it to one equation

Status: Addressed in the revised thesis

Exact correction: Population, coherence, trace, signed-cloud, source, and KDE estimators are defined once and mapped to named equations before use.

Thesis locator: Appendix E.3.1–E.3.4, printed pp. 142–144; Eqs. (E.17)–(E.33), including the signed central moments and KDE equation.

Evidence artifact: Thesis/Thesis.tex; Observables.py; final_reviewer_closure/table_data_crosswalk.csv

L-3 — Reserve evidentiary words for declared standards

Status: Addressed in the revised thesis

Exact correction: Evidentiary words are tied to declared standards: references are 'numerically controlled', independent clouds show sensitivity rather than convergence, and the method is not called validated or improved.

Thesis locator: Abstract, printed p. i; Sections 6.4–6.9, printed pp. 89–116; Sections 7.3 and 7.6, printed pp. 118–119.

Evidence artifact: Thesis/Thesis.tex; final_acceptance_check.py claim-language audit

L-4 — Correct compound-word and hyphenation defects

Status: Addressed in the revised thesis

Exact correction: The visible compounds were corrected, including 'Chapters 2–5' and 'cross-validation'; protected typography prevents line-break artefacts.

Thesis locator: Section 1.9, printed p. 25; Section 4.7.1, printed p. 60.

Evidence artifact: Thesis/Thesis.tex; reviewer_data_audit/pdf_qa/pdf_compile_render_manifest.json

L-5 — Replace vague antecedents

Status: Addressed in the revised thesis

Exact correction: Vague antecedents were replaced by named subjects, for example 'The cloud-size study in Table 6.8' and 'This conclusion applies to the tested discretization'.

Thesis locator: Section 6.5, printed p. 93; Section 7.6, printed p. 119.

Evidence artifact: Thesis/Chapter6_VerifiedResults.tex; Thesis/Chapter7_Conclusions.tex

L-6 — Compress repetitive figure narration

Status: Addressed in the revised thesis

Exact correction: Repetitive legacy figure narration was removed; five compact captions carry definitions while surrounding prose states one interpretation per figure.

Thesis locator: Sections 6.2–6.8, printed pp. 84–115; Figures 6.1–6.5.

Evidence artifact: final_reviewer_closure/figures/FIGURE_DATA_CROSSWALK.csv

L-7 — Make the response document an audit trail

Status: Addressed in the revised thesis

Exact correction: The response is now generated from a 58-row gate/item audit matrix; every row has a unique correction, exact section/page locator, and named evidence artifact.

Thesis locator: This response, Ten acceptance gates and I/M/L sections.

Evidence artifact: reviewer_data_audit/response_item_audit.csv

Final scientific statement

The revision retains unfavourable and unstable outcomes rather than turning numerical completion into a success claim. Evidence is presented as physical observables, raw invariants, operator errors, uncertainty intervals, and controlled-reference comparisons. No internal surrogate diagnostic is presented as a physical-error estimate.

Availability note

The current research-code repository is `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics`. The versioned public release is available at `https://github.com/sahandgit/gaussian_process_guided_mapping_quantumclassical_liouville_dynamics/releases/tag/thesis-final-2026-08-01-r3`. No DOI or institutional persistent identifier has been assigned.